Program 1: Orange Level

Write a C program to create and display a **singly linked list**.

The program should perform the following tasks:

1. Ask the user to enter the number of nodes to be created.
2. Dynamically create the required number of nodes using malloc().
3. Read the data value for each node from the user.
4. Link each node to the next node to form a singly linked list.
5. Use a while loop for creating the nodes.
6. Use a while loop to display the elements of the linked list.
7. Display the linked list elements in the order in which they are entered.

Sample Input:

Enter the number of nodes: 4
Enter data for node 1: 10
Enter data for node 2: 20
Enter data for node 3: 30
Enter data for node 4: 40

Sample Output:

The elements in the singly linked list are:
10 -> 20 -> 30 -> 40 -> NULL

```c
#include <stdio.h>
#include<stdlib.h>
struct Node
{
    int data;
    struct Node*next;
};

void main()
{
    int x,n, i=0;
    struct Node*newnode,*head,*temp;
    head=0;
    printf("Enter the number of nodes : ");
    scanf("%d",&n);

    while(i<n)
    {
        newnode=(struct Node*)malloc(sizeof(struct Node));
        printf("Enter the data of node %d : ",i+1);
        scanf("%d",&x);
        i++;

        newnode -> data=x;
        newnode -> next=0;

        if(head==0)
        {
            head = newnode;
            temp=head;
        }

        else
        {
            temp -> next = newnode;
            temp = newnode;
        }
    }

    temp=head;

    printf("The elements in the singly linked list are: \n");
    while(temp!=0)
    {
        printf("%d -> ",temp -> data);
        temp = temp -> next;
    }
    if(temp==0)
    {
        printf("NULL");
    }
    free(newnode);
}
```

```
prog1 }
Enter the number of nodes : 4
Enter the data of node 1 : 10
Enter the data of node 2 : 20
Enter the data of node 3 : 30
Enter the data of node 4 : 40
The elements in the singly linked list are:
10 -> 20 -> 30 -> 40 -> NULL
PS C:\Users\LENOVO\Desktop\Coding\C\weekprog.c\week12>
```

Program 2: Orange Level

Write a C program to perform insertion and deletion operations on a singly linked list.

The program should first create a singly linked list by reading data from the user. Then, perform the following operations:

1. Insert a new node at the beginning of the linked list.
2. Display the linked list after insertion at the beginning.
3. Insert a new node at the end of the linked list.
4. Display the linked list after insertion at the end.
5. Delete a node from the beginning of the linked list.
6. Display the linked list after deletion from the beginning.
7. Delete a node from the end of the linked list.
8. Display the linked list after deletion from the end.

The program should use dynamic memory allocation for creating nodes and should not use arrays.

Sample input:
Enter the number of nodes: 3



Enter data for node 1: 10
Enter data for node 2: 20
Enter data for node 3: 30

Enter value to insert at the beginning: 5
Enter value to insert at the end: 40

Sample output:
Initial linked list:
10 -> 20 -> 30 -> NULL

After inserting 5 at the beginning:
5 -> 10 -> 20 -> 30 -> NULL

After inserting 40 at the end:
5 -> 10 -> 20 -> 30 -> 40 -> NULL

After deleting a node from the beginning:
10 -> 20 -> 30 -> 40 -> NULL

After deleting a node from the end:
10 -> 20 -> 30 -> NULL

C > weekprog.c > week12 > C prog2.c > main()

```c
#include<stdio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node* next;
};

// display function
void display(struct node* head)
{
    struct node* temp = head;
    while(temp != NULL)
    {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}

int main()
{
    int n, i, x, begx, endx;
    struct node *head = NULL, *temp, *newnode, *prev;

    printf("Enter number of nodes: ");
    scanf("%d", &n);

    //Create list
    for(i = 0; i < n; i++)
    {
        newnode = (struct node*)malloc(sizeof(struct node));

        printf("Enter data for node %d: ", i+1);
        scanf("%d", &x);

        newnode->data = x;
        newnode->next = NULL;

        if(head == NULL)
        {
```

```c
            head = newnode;
            temp = newnode;
        }
        else
        {
            temp->next = newnode;
            temp = newnode;
        }
    }

    printf("\nInitial linked list:\n");
    display(head);

    //Insert at beginning
    printf("\nEnter value to insert at beginning: ");
    scanf("%d", &begx);

    newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = begx;
    newnode->next = head;
    head = newnode;

    printf("\nAfter inserting %d at the beginning:\n", begx);
    display(head);

    //Insert at end
    printf("\nEnter value to insert at end: ");
    scanf("%d", &endx);

    newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = endx;
    newnode->next = NULL;

    temp = head;
    while(temp->next != NULL)
    {
        temp = temp->next;
    }
    temp->next = newnode;

```

```c
    printf("\nAfter inserting %d at the end:\n", endx);
    display(head);

    //Delete from beginning
    if(head != NULL)
    {
        temp = head;
        head = head->next;
        free(temp);
    }

    printf("\nAfter deleting a node from the beginning:\n");
    display(head);

    //Delete from end
    if(head == NULL)
    {
        
    }
    else if(head->next == NULL)
    {
        free(head);
        head = NULL;
    }
    else
    {
        temp = head;
        while(temp->next != NULL)
        {
            prev = temp;
            temp = temp->next;
        }
        prev->next = NULL;
        free(temp);
    }

    printf("\nAfter deleting a node from the end:\n");
    display(head);

    return 0;
}
```

```
prog2 }
Enter number of nodes: 3
Enter data for node 1: 10
Enter data for node 2: 20
Enter data for node 3: 30

Initial linked list:
10 -> 20 -> 30 -> NULL

Enter value to insert at beginning: 5

After inserting 5 at the beginning:
5 -> 10 -> 20 -> 30 -> NULL

Enter value to insert at end: 40

After inserting 40 at the end:
5 -> 10 -> 20 -> 30 -> 40 -> NULL

After deleting a node from the beginning:
10 -> 20 -> 30 -> 40 -> NULL

After deleting a node from the end:
10 -> 20 -> 30 -> NULL
PS C:\Users\LENOVO\Desktop\Coding\C\weekprog.c\week12>
```